MATH MAGIC
THE IDEA

This guide is about learning to look at numbers differently.

The goal is not to memorize hundreds of formulas. It is to recognize a pattern and choose the easier path.

A good mental-math trick should do one of two things:

Make a calculation shorter. Or **make a difficult calculation easier to do.**

Every trick in this guide explains the idea behind it, when to use it, and how to apply it with a clear example.

Some tricks are useful for everyday calculation. Others are especially useful in timed questions, where finding the answer quickly matters more than writing every step on paper.

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01 - MULTIPLICATION TRICKS

Trick 1: The Base Jump

The Trick

When two numbers are close to the same round number, such as 100, use their distance from that base instead of multiplying the full numbers directly.

The amount a number is below the base is called its **shortfall**.

For example:

100 - 98 = 2

So 98 has a shortfall of 2 from 100.

This works especially well when both numbers are close to 10, 100, 1000, and other convenient bases.

Step-by-Step Example: 98 × 93

Step 1: Find the shortfall of 98 from 100.

100 - 98 = 2

Step 2: Find the shortfall of 93 from 100.

100 - 93 = 7

Step 3: Cross-subtract.

Take either number and subtract the other number's shortfall:

98 - 7 = 91

You can also do:

93 - 2 = 91

So the left part is 91.

Step 4: Multiply the shortfalls.
2 × 7 = 14
So the right part is 14.

Step 5: Put the two parts together.

Because the base is **100**, the right part must have **2 digits**.

91 | 14
= **9114**

Quick Tip

When using a base of **100**, always write the right part with **2 digits**.

For example:

98 × 97

Shortfalls:
100 - 98 = 2
100 - 97 = 3

Cross-subtract:
98 - 3 = 95

Multiply the shortfalls:
2 × 3 = 6

Write 6 as **06** because the base is 100.

95 | 06
= **9506**

So: 98 × 97 = **9506**

Trick 2: Smart Factor Multiplication

The Trick

Before multiplying two numbers directly, check whether one number can be rewritten using easier factors.

The goal is to replace a difficult multiplication with smaller operations.

For example:

75 = 3 × 25

125 = 1000 ÷ 8

50 = 100 ÷ 2

Example 1: 48 × 75

Step 1: Rewrite 75.

75 = 3 × 25

Step 2: Multiply 48 by 3.

48 × 3 = 144

Step 3: Multiply 144 by 25.

Since:

25 = 100 ÷ 4

we get:

144 × 25

= 144 × 100 ÷ 4

= 14400 ÷ 4

= 3600

Final Result

48 × 75 = **3600**

Example 2: 16 × 125

Step 1: Rewrite 125.

125 = 1000 ÷ 8

Step 2: Multiply 16 by 1000.

16 × 1000 = 16000

Step 3: Divide by 8.

16000 ÷ 8 = 2000

Final Result

16 × 125 = **2000**

Trick 3: The Half-Double Hack

The Trick

If one number is even, halve it and double the other number.

The product stays exactly the same.

Step-by-Step Example: 18 × 35

Step 1: Halve 18.

18 ÷ 2 = 9

Step 2: Double 35.

35 × 2 = 70

Step 3: Multiply the easier numbers.

9 × 70 = 630

Final Result

18 × 35 = **630**

Another Example: 25 × 48

Halve 48:

48 ÷ 2 = 24

Double 25:

25 × 2 = 50

Now:

50 × 24 = 1200

So:

25 × 48 = **1200**

When to Use It

Use this when doubling one number creates a round number or halving the other number makes it easier.

Trick 4: The 9 Magic

The Trick

To multiply by 9, multiply by 10 and subtract the original number.

Pattern

X × 9 = X × 10 - X (Where X is any number)

Example: 47 × 9

Step 1: Multiply 47 by 10.

47 × 10 = 470

Step 2: Subtract 47.

470 - 47 = 423

Final Result

47 × 9 = **423**

The Same Idea Works for 99 and 999

Because:

99 = 100 - 1

999 = 1000 - 1

Example: 436 × 999

Step 1: Multiply by 1000.

436 × 1000 = 436000

Step 2: Subtract 436.

436000 - 436 = 435564

Final Result

436 × 999 = **435564**

Trick 5: Multiply by 101

The Trick

101 is **one more than 100**:

101 = 100 + 1

So multiply by 100, then add the original number.

This is the opposite pattern of multiplying by 99, where you subtract the original number.

Example: 57 × 101

Step 1: Multiply by 100.

57 × 100 = 5700

Step 2: Add 57.

5700 + 57 = 5757

Final Result

57 × 101 = **5757**

Another Example

84 × 101

= 8400 + 84

= **8484**

Key Idea

When you see **101**, think:

One more than 100 → ×100 + original number

Trick 6: Equidistant Product Trick

The Trick

When two numbers are equally far from the same middle number, use: **Middle² - Distance²**

instead of multiplying directly.

What “Equidistant” Means

Take: 49 and 51

Both are exactly 1 away from 50.

So: 49 = 50 - 1

51 = 50 + 1

Step-by-Step Example: 49 × 51

Step 1: Find the middle number.

Middle = 50

Step 2: Find the distance.

Distance = 1

Step 3: Square the middle.

$50^2 = 2500$

Step 4: Square the distance.

$1^2 = 1$

Step 5: Subtract.

2500 - 1 = 2499

Final Result

49 × 51 = **2499**

Another Example: 63 × 67

Middle = 65 and Distance = 2

$65^2 = 4225$

$2^2 = 4$

4225 - 4 = 4221

Final Result

63 × 67 = **4221**

When to Use It

Look for pairs such as:

48 and 52, 74 and 76, 97 and 103, 198 and 202

The numbers must be equally far from the same middle.

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02 - DIVISION TRICKS

Trick 1: Turn the Divisor Into a Power of 10

The Trick

When a divisor is awkward to divide by, look for a number you can multiply it by to turn it into **10, 100, 1000, or another power of 10**.

Then:

Multiply the number by that same multiplier, and divide by the new power of 10.

The General Pattern

If: **Divisor × Multiplier = 10^n**

Then: **Number ÷ Divisor = (Number × Multiplier) ÷ 10^n**

This works because you are multiplying and dividing by the same value in a different form.

Example 1: 85 ÷ 5

5 × 2 = 10

So: **÷ 5 → × 2, then ÷ 10**

Step 1: Multiply by 2.

85 × 2 = 170

Step 2: Divide by 10.

170 ÷ 10 = 17

Final Result

85 ÷ 5 = **17**

Example 2: 312 ÷ 25

25 × 4 = 100
So: **÷ 25 → × 4, then ÷ 100**

Step 1: Multiply by 4.

312 × 4 = 1248

Step 2: Divide by 100.

1248 ÷ 100 = 12.48

Final Result

312 ÷ 25 = **12.48**

Example 3: 4375 ÷ 125

125 × 8 = 1000
So: **÷ 125 → × 8, then ÷ 1000**

Step 1: Multiply by 8.

4375 × 8 = 35000

Step 2: Divide by 1000.

35000 ÷ 1000 = 35

Final Result

4375 ÷ 125 = **35**

Example 4: 840 ÷ 20

20 × 5 = 100
So: **÷ 20 → × 5, then ÷ 100**

Step 1: Multiply by 5.

840 × 5 = 4200

Step 2: Divide by 100.

4200 ÷ 100 = 42

Final Result

840 ÷ 20 = **42**

Example 5: 960 ÷ 40

40 × 25 = 1000

So: **÷ 40 → × 25, then ÷ 1000**

Step 1: Multiply by 25.

960 × 25 = 24000

Step 2: Divide by 1000.

24000 ÷ 1000 = 24

Final Result

960 ÷ 40 = **24**

Trick 2: Factor-First Division

The Trick

When a divisor looks difficult, break it into smaller factors that are easier to divide by.

Instead of dividing directly, divide step by step.

For example:

36 = 9 × 4

So:

÷ 36 → ÷ 9 → ÷ 4

This works especially well when the divisor can be broken into familiar numbers such as **2, 3, 4, 5, 8, 9, 10,** and so on.

Example 1: 567 ÷ 9

Break 9 into:

9 = 3 × 3

So: **÷ 9 → ÷ 3 → ÷ 3**

Step 1: Divide by 3.

567 ÷ 3 = 189

Step 2: Divide by 3 again.

189 ÷ 3 = 63

Final Result

567 ÷ 9 = **63**

Example 2: 648 ÷ 36

Break 36 into: **36 = 9 × 4**

So: **÷ 36 → ÷ 9 → ÷ 4**

Step 1: Divide by 9.

648 ÷ 9 = 72

Step 2: Divide by 4.

72 ÷ 4 = 18

Final Result

648 ÷ 36 = **18**

Quick Tip

Before doing long division, ask:

Can I break the divisor into smaller, easier factors?

For example:

÷ 9 → ÷ 3 → ÷ 3

÷ 36 → ÷ 9 → ÷ 4

÷ 16 → ÷ 2 → ÷ 2 → ÷ 2 → ÷ 2

Trick 3: Divide by 11 Using Known Multiples

The Trick

When dividing by 11, break the number into smaller parts that are easy multiples of 11.

Instead of doing long division, divide each part separately and add the results.

Example: 1210 ÷ 11

Step 1: Split 1210 into easy parts.

1210 = 1100 + 110

Step 2: Divide both parts by 11.

(1100 + 110) ÷ 11

= 1100 ÷ 11 + 110 ÷ 11

Step 3: Use the known multiples.

1100 ÷ 11 = 100

110 ÷ 11 = 10

Step 4: Add the results.

100 + 10 = 110

Final Result

1210 ÷ 11 = **110**

Another Example: 1430 ÷ 11

Step 1: Split the number.

1430 = 1100 + 330

Step 2: Divide each part.

(1100 + 330) ÷ 11

= 1100 ÷ 11 + 330 ÷ 11

= 100 + 30

Step 3: Add.

100 + 30 = 130

Final Result

1430 ÷ 11 = **130**

Trick 4: The Ugly Division Killer: Repeated Halving

The Trick

When the divisor is a power of 2, keep halving the number.

Instead of dividing by 16 in one difficult step: ÷ 16

you can do: ÷ 2 → ÷ 2 → ÷ 2 → ÷ 2

Example: 960 ÷ 16

Step 1: Divide by 2.

960 ÷ 2 = 480

Step 2: Divide by 2 again.

480 ÷ 2 = 240

Step 3: Divide by 2 again.

240 ÷ 2 = 120

Step 4: Divide by 2 one more time.

120 ÷ 2 = 60

Final Result

960 ÷ 16 = **60**

Key Idea

This is especially useful for:

÷ 4

÷ 8

÷ 16

÷ 32

÷ 64

You do not have to attack the ugly divisor directly.

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03 - PERCENTAGE & FRACTION TRICKS

Trick 1: Build Any Percentage

The Trick

Break the target percentage into **easy percentages you already know**.

Useful building blocks include:

10%, 5%, 1%, 2%, 0.5%, 20%, 25%, 50%

Choose the combination that makes the calculation easiest.

Step-by-Step Example: 35% of 640

This is the new pattern to remember:

35% = 30% + 5%

Step 1: Find 10%.

10% of 640 = 64

Step 2: Build 30%.

64 × 3 = 192

So:

30% of 640 = 192

Step 3: Find 5%.

5% is half of 10%.

64 ÷ 2 = 32

Step 4: Add the two parts.

192 + 32 = 224

Final Result

35% of 640 = **224**

Another Example: 17.5% of 640

17.5% = 10% + 5% + 2% + 0.5%

10% = 64
5% = 32
2% = 12.8
0.5% = 3.2

Add: 64 + 32 + 12.8 + 3.2 = 112

Final Result

17.5% of 640 = **112**

Key Idea

A percentage does not always need to be calculated as a decimal.

Build it from percentages you already know.

Trick 2: The Percentage-Fraction Shortcut

The Trick

Some percentages are actually very simple fractions.

Memorize these:

Percentage	Fraction
50%	1/2
25%	1/4
12.5%	1/8
5%	1/20
2.5%	1/40
37.5%	3/8
62.5%	5/8
75%	3/4

Example: 37.5% of 480

Instead of multiplying by 0.375, use:

37.5% = 3/8

Step 1: Divide 480 by 8.

480 ÷ 8 = 60

Step 2: Multiply by 3.

60 × 3 = 180

Final Result

37.5% of 480 = **180**

Example: 62.5% of 88

62.5% = 5/8

88 ÷ 8 = 11

11 × 5 = 55

Final Result

62.5% of 88 = **55**

Trick 3: Percentage of a Percentage

The Trick

When finding one percentage of another percentage, **multiply the two percentage numbers and divide by 100**.

Pattern:
A% of B% = (A × B ÷ 100)%

Here:
A = first percentage
B = second percentage

Example: 20% of 30% of 500

Here:
A = 20
B = 30

First combine the percentages:

20 × 30 = 600
600 ÷ 100 = 6%

Now find **6% of 500**:

6% of 500 = 30

Final Result

20% of 30% of 500 = 30

Trick 4: Percentage Reversal

The Trick

You can switch the number and percentage: **A% of B = B% of A**

Choose whichever direction is easier.

Example: 84% of 50

84% of 50 = 50% of 84
50% means half.
84 ÷ 2 = 42

Final Result

84% of 50 = **42**

Another Example

25% of 80 = 80% of 25

80% of 25 = 20

So: 25% of 80 = **20**

Trick 5: Find 1% First

The Trick

When the target percentage is awkward, find **1% first** and then multiply it by the required percentage.

1% = Number ÷ 100

This is useful when breaking the percentage into several familiar parts would take more steps.

Example: 7% of 850

Step 1: Find 1%.

850 ÷ 100 = 8.5

Step 2: Multiply by 7.

8.5 × 7 = 59.5

Final Result

7% of 850 = **59.5**

Trick 6: The Reverse Percentage Trick

The Trick

If you know the final value after an increase or decrease, do not subtract the percentage directly. Instead, work backward using the final multiplier.

Example: A price becomes 600 after a 20% increase.

Step 1: Understand the new percentage.

Original = 100%

Increase = 20%

Final = 120%

120% = 1.2

Step 2: Divide the final amount by 1.2.

600 ÷ 1.2 = 500

Final Result

Original Price = **500**

Check

500 × 1.2 = 600

So the answer is correct.

Trick 7: Successive Percentage Changes

The Trick

Two percentage changes do not simply cancel or add. The second percentage is applied to the already changed amount.

Example: +20% followed by -20%

Start with 100.

Step 1: Increase by 20%.

100 → 120

Step 2: Decrease the new amount by 20%.

20% of 120 = 24
120 - 24 = 96

Final Result

100 → 96
So the overall result is a: **4% decrease**

Why They Do Not Cancel

The first 20% was taken from 100.
The second 20% was taken from 120.
Those two 20% amounts are not equal.

Trick 8: Fraction Comparison Without Division

The Trick

To compare two fractions, cross-multiply.

You do not need to convert either fraction into a decimal.

Example: Compare 5/7 and 7/10

Step 1: Multiply the first numerator by the second denominator.

5 × 10 = 50

Step 2: Multiply the second numerator by the first denominator.

7 × 7 = 49

Step 3: Compare.

50 > 49

Therefore:

5/7 > 7/10

Final Result

5/7 is greater than 7/10

Trick 9: Fraction Multiplication by Cancellation

The Trick

Before multiplying fractions, look for numbers that can cancel.

Example

3/4 × 8/9

Instead of multiplying first: 3 × 8 = 24 Or 4 × 9 = 36

Simplify 24/36 later.

Cancel first.

3 and 9 can be divided by 3:

3 → 1

9 → 3

8 and 4 can be divided by 4:

8 → 2

4 → 1

Now:

1/1 × 2/3

= 2/3

Final Result

3/4 × 8/9 = **2/3**

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04 - Ultimate Formula Shortcuts

Quick Vocabury

Cost Price (CP) = price paid to buy an item.

Selling Price (SP) = price received when an item is sold.

Marked Price (MP) = price shown before discount.

Markup = the percentage added to the Cost Price to set the Marked Price.

Discount = the percentage reduced from the Marked Price to get the Selling Price.

Trick 1: Combine Successive Profit & Loss

When to Use It

Use this trick when the **same item goes through two percentage changes one after another**, such as:

20% Profit → 10% Loss
Or
20% Profit → 15% Profit

Do not simply add or subtract the percentages.

The Trick

Treat a profit as **+** and a loss as **−**.

Net Change = A + B + (A × B ÷ 100) **(**Here, **A** and **B** are the two percentage changes**).**

Example: 20% Profit, Then 10% Loss

A = +20
B = −10

20 − 10 + (20 × −10 ÷ 100)
= 10 − 2
= 8%

Result: 8% Profit

Remember

The second percentage is applied to the **new value**, not the original value.

Trick 2: Same Selling Price, Equal Profit & Loss

When to Use It

Use this trick when **two items are sold at the same Selling Price**, with one earning **x% Profit** and the other incurring **x% Loss**.

The Trick

Instead of finding both Cost Prices separately, square the percentage and divide by 100.

Net Loss = $x^2 \div 100$ %

Here, **x** is the equal profit/loss percentage.

Example: 20% Profit and 20% Loss

$20^2 \div 100$
= 400 ÷ 100
= 4%

Result: 4% Loss

Remember
Same SP + Same % Profit/Loss → Square the % → ÷ 100

Trick 3: False Weight Trick

When to Use It

Use this trick when a seller:

- Charges the price of the **full quantity**
- But actually gives **less quantity**

The Trick

Find the missing quantity, then compare it with the **quantity actually given**.

Gain % = (True Quantity − Actual Quantity) ÷ Actual Quantity × 100

Example: 900 g Given Instead of 1 kg

True Quantity = 1000 g
Actual Quantity = 900 g

Missing quantity:

1000 − 900 = 100 g

Now:

100 ÷ 900 × 100
= 11.11%

Result: 11.11% Gain

Remember
The denominator is the **quantity actually given**.

Trick 4: Markup + Discount

When to Use It

Use this trick when an item is:

Marked up by M% → Then sold at D% Discount

You can find the final profit directly without calculating the Marked Price and Selling Price separately.

The Trick

Net Profit = M − D − (M × D ÷ 100)

Here:

M = Markup %
D = Discount %

Example: 30% Markup, 10% Discount

30 − 10 − (30 × 10 ÷ 100)
= 20 − 3
= 17%

Result: 17% Profit

Remember

The discount is taken **after the markup**, so the two percentages cannot simply be subtracted.

Trick 5: Successive Discounts

When to Use It

Use this trick when an item gets **two discounts one after another**.

For example: **20% Discount → 10% Discount**

The Trick

Net Discount = A + B − (A × B ÷ 100)

Here, **A** and **B** are the two discount percentages.

Example: 20% and 10% Discounts

20 + 10 − (20 × 10 ÷ 100)
= 30 − 2
= 28%

Result: 28% Discount

Remember
20% + 10% ≠ 30%
The second 10% is taken from the **already reduced price**.

Trick 6: Rule of 72

When to Use It

Use this trick when you want to quickly estimate **how many years it takes money to double** at a given annual interest rate.

The Standard Approach

Without this trick, finding the doubling time exactly involves using the **compound interest formula**, which is more difficult to calculate mentally.

Compound Amount = $P \times (1 + r/100)^t$

Where:

P = the starting amount of money
r = the annual interest rate (%)
t = the number of years

This formula assumes the interest is **compounded annually**.

To find the doubling time, you need to work out when the final amount becomes **2 × the starting amount**.

This requires extra calculations, making it inconvenient for quick mental estimation.

The Trick

Doubling Time ≈ 72 ÷ Interest Rate

Here, **Interest Rate** is the annual interest rate in percentage.

Example: 8% Interest

72 ÷ 8
= 9 years

Result: Your money will roughly **double in 9 years**.

Remember

72 ÷ Interest Rate → Approximate Years to Double

This is a **quick estimation rule**, not an exact calculation.

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05 - DIVISIBILITY RULES

Use this section as a quick-reference table. The goal is to check divisibility without performing the full division.

Number	Divisibility Rule	Example
2	Last digit is even	438 ends in 8, so it is divisible by 2
3	Digit sum is divisible by 3	4572 → 4+5+7+2 = 18
4	Last two digits are divisible by 4	5316 → 16 is divisible by 4
5	Last digit is 0 or 5	735 ends in 5
6	Divisible by both 2 and 3	1236 is even and digit sum is 12
7	Double the last digit and subtract it from the remaining number. Repeat if necessary.	203 → 20 - 6 = 14
8	Last three digits are divisible by 8	12824 → 824 ÷ 8 = 103
9	Digit sum is divisible by 9	7281 → 7+2+8+1 = 18
10	Last digit is 0	450 ends in 0
11	Difference between alternating digit sums is 0 or a multiple of 11	48273 → (4+2+3) - (8+7) = -6, so not divisible
12	Divisible by both 3 and 4	1236 passes both tests

Quick Memory Pattern

For many divisibility questions, you do not need the entire number.

Look at:
Last digit → 2, 5, 10
Last two digits → 4
Last three digits → 8
Digit sum → 3, 9
Double last digit and subtract → 7
Two separate checks → 6, 12

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06 - FAST CALCULATION SHORTCUTS

These methods help you **calculate faster by changing the way you work with numbers**. They are especially useful when a calculation looks longer than it really is.

Shortcut 1: Round and Correct

The Idea

Replace a difficult number with a nearby round number, calculate quickly, and then correct the difference.

Example 1: 47 × 6

Step 1: Round 47 to 50.

50 × 6 = 300

Step 2: Find the difference.

50 - 47 = 3

You added 3 too much.

Step 3: Correct the result.

3 × 6 = 18

300 - 18 = 282

Final Result

47 × 6 = **282**

Example 2: 800 - 397

Step 1: Round 397 to 400.

800 - 400 = 400

Step 2: Find the difference.

400 - 397 = 3
You subtracted 3 too much.

Step 3: Correct the result.

400 + 3 = 403

Final Result

800 - 397 = **403**

Quick Tip

Use this when a number is close to an easy round number such as: **10, 20, 50, 100, 200, 500, 1000**
Round first. Then always ask:

Did I add too much or subtract too much?

Shortcut 2: Group Numbers to Make 100

The Idea

When adding several numbers, look for pairs or groups that create **100, 200, 500, or another easy total**.

Example

48 + 52 + 27 + 73

Step 1: Make 100.

48 + 52 = 100

Step 2: Make another 100.

27 + 73 = 100

Step 3: Add the groups.

100 + 100 = 200

Final Result

200

Shortcut 3: Subtract From 1000

The Idea

For a three-digit number subtracted from 1000:

- Subtract each earlier digit from 9.
- Subtract the final digit from 10.

Example: 1000 - 347

Step 1:

9 - 3 = 6

Step 2:

9 - 4 = 5

Step 3:

10 - 7 = 3

Put the digits together: **653**

Final Result

1000 - 347 = **653**

Another Example

1000 - 528

9 - 5 = 4
9 - 2 = 7
10 - 8 = 2

Answer = **472**

Quick Reminder

For: **1000 - ABC**

Think: **9 - A, 9 - B, 10 - C**

This makes the calculation almost automatic.

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07 - Time-Saving Approaches

The methods in this section are **not calculation tricks**.

They are approaches you can use to:

- estimate an answer
- eliminate wrong options
- check a calculation
- avoid unnecessary work

The goal is simple:

Do not calculate more than the question requires.

Approach 1: E — Estimation

What It Does

Estimation gives you a **rough target** before you calculate the exact answer.

It helps you quickly notice when an answer is far too large or too small.

When to Use It

Use estimation when:

- numbers are close to round numbers
- you want to check the size of an answer
- the exact calculation is lengthy
- you want a quick error check

Example: 398 × 21

Step 1: Estimate.

398 is close to 400.

400 × 21 = 8400

So the exact answer should be somewhere near **8400**.

Step 2: Calculate exactly.

398 × 21 = 398 × 20 + 398
= 7960 + 398
= 8358

Final Result

398 × 21 = **8358**

Since 8358 is close to 8400, the answer passes the estimation check.

Important

Estimation does **not** prove that the answer is correct.

It only tells you whether the answer is in the expected range.

Approach 2: OE — Option Elimination

What It Does

In multiple-choice questions, you often do not need the complete calculation.

Instead, identify a property that the correct answer **must have** and eliminate the options that fail.

When to Use It

Use this approach when:

- there are multiple options
- one quick check can remove several options
- you only need to identify the correct choice
- doing the full calculation would waste time

Example: 47 × 23

Options:

A. 1002
B. 1081
C. 1090
D. 1115

Step 1: Check the last digit.

7 × 3 = 21

So the answer must end in **1**.

Only **1081** ends in 1.

The other three options can be eliminated immediately.

Step 2: Check the approximate size.

47 × 23 is roughly:

50 × 20 = 1000

1081 is in the expected range.

Final Result

1081

You found the answer without performing the full multiplication.

Key Idea

In an MCQ, ask:

What is the minimum information I need to identify the answer?

Sometimes one digit is enough.

Approach 3: DRC — Digital Root Check

What It Does

A digital root reduces a number to a single digit by repeatedly adding its digits.

It is useful for checking whether a calculated answer **could be correct**.

When to Use It

Use digital roots when:

- checking multiplication or addition
- verifying a long calculation
- you want a second check after calculating
- you want a quick mental verification

Example: Verify 43 × 21 = 903

Step 1: Find the digital root of 43.

4 + 3 = 7

Digital root = **7**

Step 2: Find the digital root of 21.

2 + 1 = 3

Digital root = **3**

Step 3: Multiply the digital roots.

7 × 3 = 21

2 + 1 = 3

Expected digital root = **3**

Step 4: Check 903.

9 + 0 + 3 = 12

1 + 2 = 3

The answer also has digital root **3**.

Final Result

The calculation **passes the digital-root check**.

Important

A matching digital root does **not prove** that the answer is correct.

It only tells you that the answer has passed this particular check.

A wrong answer can sometimes have the same digital root.

Approach 4: MC — Multi-Check

What It Does

Instead of relying on one check, combine several quick checks.

For example:

Unit Digit + Digital Root + Magnitude

Each check looks at the answer from a different angle.

When to Use It

Use a multi-check when:

- the calculation is important
- you want stronger confidence
- the numbers are large
- you suspect a small calculation mistake
- you want to verify an answer quickly without recalculating everything

Example: 347 × 68 = 23596

Check 1: Unit Digit

7 × 8 = 56

So the answer must end in **6**.

23596 ends in 6.

Pass.

Check 2: Digital Root

347: 3 + 4 + 7 = 14 & 1 + 4 = 5

68:
6 + 8 = 14 & 1 + 4 = 5

5 × 5 = 25 & 2 + 5 = 7

Now check 23596:
2 + 3 + 5 + 9 + 6 = 25 & 2 + 5 = 7

Pass.

Check 3: Magnitude

350 × 70 ≈ 24500

23596 is in the same general range.

Pass.

Final Result

All three checks agree: 347 × 68 = **23596**

---

08 - SQUARE & SQUARE ROOT TRICKS

Trick 1: Square Using a Nearby Base

The Trick

Choose a convenient number close to the number you want to square.

Then use the difference from that base.

This is the main base method for squaring.

Example: 47^2

Choose:

Base = 50

Difference:

47 - 50 = -3

Step 1: Square the base.

$50^2 = 2500$

Step 2: Calculate twice the base times the difference.

$2 \times 50 \times (-3) = -300$

Step 3: Square the difference.

$(-3)^2 = 9$

Step 4: Combine everything.

2500 - 300 + 9 = 2209

Final Result

47^2 = **2209**

Important

You do not need to memorize separate tricks for 47^2, 48^2, 52^2, and so on.
They are all applications of the same base idea.

Trick 2: Square Numbers Ending in 5

The Trick

To square any number that ends in 5 (**N5**), multiply the remaining leading digit(s) (**N**) by the next consecutive whole number (**N + 1**), then place **25** at the end of the product.

Pattern: [N × (N + 1)] attached with 25

Step-by-Step Example: Calculate 35^2

1. **Identify the leading digit (N):**
   For 35, the digit before 5 is N = 3.
2. **Find the next consecutive whole number (N + 1):**
   3 + 1 = 4
3. **Multiply the leading digit by the next number:**
   3 × 4 = 12
4. **Append 25 to the result:**
   Attach 25 directly to the right of 12 → **1225**.
5. **Final Result:**
   35^2 = **1225**

Trick 3: Find the Next Square

The Trick

If you know the square of a number, use it to quickly find the square of the number that comes after it.

How to Use It

Use this simple rule: **Next Square = Known Square + (2 × Number) + 1**

Example: Find 41^2 if $40^2 = 1600$

Step 1: Double 40.
2 × 40 = 80

Step 2: Add 1.
80 + 1 = 81

Step 3: Add 81 to the known square.
1600 + 81 = 1681

Final Result

41^2 = **1681**

Another Example

If:

$50^2 = 2500$

To find 51^2:

2 × 50 = 100
100 + 1 = 101
2500 + 101 = **2601**

So:

51^2 = **2601**

Trick 4: Square Numbers From 11 to 19

The Trick

For numbers from **11 to 19**, you can use a quick shortcut based on the units digit. This is especially useful when you want to square a number in this range quickly without using the longer general squaring method.

Example: 14^2

Units digit = 4

Step 1: Add the units digit to the original number.

14 + 4 = 18

Step 2: Multiply by 10.

18 × 10 = 180

Step 3: Square the units digit.

$4^2 = 16$

Step 4: Add.

180 + 16 = 196

Final Result

14^2 = **196**

Another Example: 17^2

Units digit = 7

17 + 7 = 24

24 × 10 = 240

$7^2 = 49$

240 + 49 = 289

Final Result

17^2 = **289**

Trick 5: Perfect Square Last-Digit Check

The Trick

A square number cannot end in:

2, 3, 7, or 8

So these digits let you reject a number instantly.

Example: Is 1237 a perfect square?

Step 1: Look at the final digit.

1237 ends in 7.

Step 2: Check the allowed square endings.

7 is impossible.

Final Result

1237 is **not** a perfect square.

Important

An allowed ending does not prove that a number is a perfect square.
For example, 24 ends in 4, but 24 is not a perfect square.
This trick is an elimination check.

Trick 6: Estimate a Square Root

The Trick

Find the two nearby perfect squares surrounding the number.

Example: √150

Step 1: Find the nearby squares.
$12^2 = 144$
$13^2 = 169$
So: $12 < \sqrt{150} < 13$

Step 2: Find how far 150 is above 144.
150 - 144 = 6

Step 3: Find the total gap.
169 - 144 = 25

Step 4: Estimate the position inside the gap.
6 ÷ 25 = 0.24

Step 5: Add to the lower root.
12 + 0.24 = 12.24

Final Result
$\sqrt{150} \approx$ **12.24**

Important
This is an estimate, not an exact answer.

Trick 7: Digit-by-Digit Squaring

The Trick

For a two-digit number, split it into its **tens part** and **units part**.

Then:

1. Square the tens part.
2. Multiply the tens part by **twice the units digit**.
3. Square the units digit.
4. Add all three results.

Example: 43^2

$43 = 40 + 3$

Step 1: Square the tens part.

$40^2 = 1600$

Step 2: Multiply the tens part by twice the units digit.

The units digit is 3.

Twice 3 = 6

Now:

$40 \times 6 = 240$

Step 3: Square the units digit.

$3^2 = 9$

Step 4: Add the three results.

$1600 + 240 + 9 = 1849$

Final Result

$43^2 =$ **1849**

Another Example: 27^2

$27 = 20 + 7$

Step 1: Square the tens part.

$20^2 = 400$

Step 2: Double the units digit.

$7 \times 2 = 14$

Now multiply by the tens part:

$20 \times 14 = 280$

Step 3: Square the units digit.

$7^2 = 49$

Step 4: Add.
400 + 280 + 49 = 729

Final Result
27^2 = **729**

---

09 - CUBE & CUBE ROOT TRICKS

Trick 1: Cube Using a Nearby Base

The Trick

When a number is close to a convenient base such as 10, 20, 50, or 100, use the base to make the cube easier.

Instead of multiplying the full number three times, write the number as:

Base + Difference

Then work with the smaller difference.

Example: 11^3

11 is only 1 above 10.

So: 11 = 10 + 1

We now have: $11^3 = (10 + 1)^3$

Why Do the 3s Appear?

You may notice that the calculation contains **3 ×** in two places.

That happens because we are multiplying three copies of: **(10 + 1)**

When the brackets are multiplied, the terms containing: 10 × 10 × 1
appear three times:

10 × 10 × 1
10 × 1 × 10
1 × 10 × 10

All three give the same result:

100

So together they give:

100 + 100 + 100 = 300

The same thing happens with:

10 × 1 × 1

1 × 10 × 1

1 × 1 × 10

Each gives 10, so together they give:

10 + 10 + 10 = 30

That is where the two **3s** come from.

Step-by-Step Calculation

Step 1: Cube the base.

$10^3 = 1000$

Step 2: Calculate the three copies of 10 × 10 × 1.

$3 \times 10^2 \times 1$

$= 3 \times 100$

$= 300$

Step 3: Calculate the three copies of 10 × 1 × 1.

$3 \times 10 \times 1^2$

$= 3 \times 10$

$= 30$

Step 4: Cube the difference.

$1^3 = 1$

Step 5: Add everything.

1000 + 300 + 30 + 1 = 1331

Final Result

11^3 = **1331**

Example: 9^3

This time the number is 1 below 10.

So: 9 = 10 - 1

The same idea works, but the negative difference changes the signs.

Step 1: Cube the base.

$10^3 = 1000$

Step 2: Subtract the three copies of 10 × 10 × 1.

$3 \times 10^2 \times 1 = 300$

1000 - 300 = 700

Step 3: Add the three copies of 10 × 1 × 1.

$3 \times 10 \times 1^2 = 30$

700 + 30 = 730

Step 4: Subtract the cube of the difference.

$1^3 = 1$

730 - 1 = 729

Final Result

9^3 = **729**

Trick 2: Cube a Number Ending in 5

The Trick

If a number ends in **5**, split it into:

5 × the number before 5

Then: $\mathbf{5^3 = 125}$

So: $\mathbf{(5 \times N)^3 = 125 \times N^3}$

And: **×125 = ×1000 ÷ 8**

This can make the cube easier to calculate.

Example: 35^3

Step 1: Rewrite 35.

$35 = 5 \times 7$

So: $35^3 = 5^3 \times 7^3$

Step 2: Cube 5.

$5^3 = 125$

So: $35^3 = 125 \times 7^3$

Step 3: Cube 7.

$7^3 = 343$

So: $35^3 = 125 \times 343$

Step 4: Multiply by 125 using the power-of-10 shortcut.

$343 \times 125 = 343 \times 1000 \div 8$
$= 343000 \div 8 = 42875$

Final Result

35^3 = **42875**

Example: 25^3

Step 1: Rewrite 25.

$25 = 5 \times 5$

Step 2: Cube both parts.

$5^3 \times 5^3 = 125 \times 125$

Step 3: Calculate.

$125 \times 125 = 15625$

Final Result

25^3 = **15625**

Key Idea

When a number ends in 5, look at the number after removing the 5.

For example:

15 = 5 × 3

25 = 5 × 5

35 = 5 × 7

45 = 5 × 9

Then use: **$5^3 = 125$**

This can turn an awkward cube into a smaller familiar cube plus a ×125 calculation.

Trick 3: Build the Next Cube From a Known Cube

The Trick

When you already know the cube of a number, use it to quickly find the cube of the **next number**.

For a known number **n**:

Next Cube = Known Cube + 3 × n^2 + 3 × n + 1

This works for consecutive numbers.

This is useful when you already know common cubes such as:

$10^3 = 1000$

$20^3 = 8000$

Instead of starting the cube calculation from scratch, build the next cube from the one you already know.

Example: Find 11^3 From 10^3

Start with: $10^3 = 1000$

Step 1: Square 10.
$10^2 = 100$

Step 2: Multiply the square by 3.
3 × 100 = 300

Step 3: Multiply 10 by 3.
3 × 10 = 30

Step 4: Add 1.
1

Step 5: Add everything to 1000.
1000 + 300 + 30 + 1 = 1331

Final Result
11^3 = **1331**

Example: Find 21^3 From 20^3

Start with: $20^3 = 8000$

Step 1: Square 20.
$20^2 = 400$

Step 2: Multiply by 3.
400 × 3 = 1200

Step 3: Multiply 20 by 3.
20 × 3 = 60

Step 4: Add 1.
1

Step 5: Add to the known cube.
8000 + 1200 + 60 + 1
= 9261

Final Result
21^3 = **9261**

Key Idea

Use this when you already know a cube and want to find the cube of the **next consecutive number**.

Trick 4: Find a Cube Root Quickly

The Trick

For a **perfect cube**, use its last digit to find the last digit of the cube root.

Then look at the digits before the final three digits to find the first part of the root.

Last-Digit Reference

Cube ends in	Root ends in
0	0
1	1
2	8
3	7
4	4
5	5
6	6
7	3
8	2
9	9

The useful pairs to remember are:
2 ↔ 8
3 ↔ 7

The other digits stay the same.

Example: $\sqrt[3]{2197}$

Step 1: Look at the final digit.

2197 ends in 7.

A cube ending in 7 has a cube root ending in 3.

So the root ends in: **3**

Step 2: Separate the final three digits.

2197

→ 2 | 197

Look only at the left part for the moment:

2

Step 3: Find the largest cube that fits into 2.

$1^3 = 1$
$2^3 = 8$, which is too large.

So the first part of the root is:

1

Step 4: Combine the two parts.

First part = 1
Last digit = 3

So:

$\sqrt[3]{2197}$ = **13**

Final Result

$\sqrt[3]{2197}$ = **13**

Another Example: $\sqrt[3]{2744}$

Step 1: Look at the final digit.

2744 ends in 4.

So the root also ends in 4.

Step 2: Separate the last three digits.

2744

→ 2 | 744

Left part = 2

Step 3: Find the largest cube that fits into 2.

$1^3 = 1$

$2^3 = 8$, which is too large.

So the first digit is 1.

Step 4: Combine.

1 and 4

= 14

Final Result

$\sqrt[3]{2744}$ = **14**

Important

Keep the final group as exactly three digits when using this method.

For example:

1000

should be viewed as:

1 | 000

The zeros are part of the grouping.

---

10 - TABLE TRICKS

Trick 1: Build a Difficult Table From an Easy Table

The Trick

If a difficult table is a multiple of an easier table, calculate the easier one and scale it.

Example: 8 × 7

Instead of recalling 8 × 7 directly:

Step 1: Use the 4-times table.

4 × 7 = 28

Step 2: Notice that 8 is double 4.

So double 28:

28 × 2 = 56

Final Result

8 × 7 = **56**

Another Example: 6 × 8

Use:

3 × 8 = 24

Double:

24 × 2 = 48

Final Result

6 × 8 = **48**

Trick 2: Build 11-19 Tables From 10

The Trick

For numbers from **11 to 19**, start with the **10-times table** and then add the extra part.

For example:

13 = 10 + 3

So: **13 × 7 = (10 × 7) + (3 × 7)**

This is a quick way to rebuild a table from a familiar **10-times table**.

Example: 13 × 7

13 = 10 + 3

Step 1: Multiply the 10.

10 × 7 = 70

Step 2: Multiply the extra 3.

3 × 7 = 21

Step 3: Add.

70 + 21 = 91

Final Result

13 × 7 = **91**

Another Example: 17 × 6

17 = 10 + 7

10 × 6 = 60

7 × 6 = 42

60 + 42 = 102

Final Result

17 × 6 = **102**

Trick 3: Table Trick: Rebuild Any Table You Forget

The Rule

Forgot a multiplication table? **Split the number into digits, multiply each digit by the multiplier, then handle the right side and combine the results.**

How to Use the Method

Suppose you want to find: **93 × 7**

Step 1: Split the Number

Write the digits separately:

93 → 9 | 3

Step 2: Multiply Both Digits by 7

9 × 7 = 63
3 × 7 = 21

Now write them together:

63 | 21

Step 3: Check the Right Side

Check whether the right-side result is **greater than 9**.

- **If it is 9 or less:** keep it as it is.
- **If it is greater than 9:** write its last digit on the right and carry the remaining digit(s) to the left.

Apply It Here

The right-side result is:

21

Since **21 is greater than 9**, write **1** on the right and carry **2** to the left.

63 | 21 → 63 + 2 | 1

Step 4: Add the Carry

Add the carried **2** to the left side:

63 + 2 = 65

So now: **65 | 1**

Step 5: Combine the Parts

65 | 1 = 651

Final Answer

93 × 7 = 651

Rebuilding the Complete 93 Table

Calculation	Split & Multiply	Split Result	Final Answer
93 × 1	**9|3 × 1**	**9|3**	**93**
93 × 2	**9|3 × 2**	**18|6**	**186**
93 × 3	**9|3 × 3**	**27|9**	**279**
93 × 4	**9|3 × 4**	**36|12**	**372**
93 × 5	**9|3 × 5**	**45|15**	**465**
93 × 6	**9|3 × 6**	**54|18**	**558**
93 × 7	**9|3 × 7**	**63|21**	**651**
93 × 8	**9|3 × 8**	**72|24**	**744**
93 × 9	**9|3 × 9**	**81|27**	**837**
93 × 10	**9|3 × 10**	**90|30**	**930**

---

11 - MATH MAGIC TRICKS

This is where the numbers stop feeling like ordinary calculations.

These tricks are designed to create a surprising result while the underlying mathematics stays simple enough to understand.

Magic Trick 1: Predict the Total

The Effect

You can predict the exact total after seeing just the first number, before the remaining four numbers are written.

Step-by-Step Execution

1. **Step 1: Ask a volunteer to write any 5-digit number.**
    - Suppose they write: **12345**
2. **Step 2: Calculate and write your secret prediction.**
    - Subtract **2** from the last digit: 5 - 2 = 3 → 12,343
    - Place a **2** at the very front: **212343**
    - Write **212343** on a piece of paper and hide it face-down.
3. **Step 3: Ask the volunteer to write a second 5-digit number.**
    - Suppose they write: **23456**
4. **Step 4: Write the third number yourself using the "Make 9" rule.**
    - Choose each digit so that it adds up to **9** when paired with the volunteer's second digit:
        - 9 - 2 = 7
        - 9 - 3 = 6
        - 9 - 4 = 5
        - 9 - 5 = 4
        - 9 - 6 = 3
    - You write: **76543** *(Notice: 23456 + 76543 = 99999)*
5. **Step 5: Ask the volunteer to write a fourth 5-digit number.**
    - Suppose they write: **54321**
6. **Step 6: Write the fifth number yourself using the same "Make 9" rule.**
    - Pair each digit with the volunteer's fourth digit to total **9**:
        - 9 - 5 = 4
        - 9 - 4 = 5
        - 9 - 3 = 6
        - 9 - 2 = 7
        - 9 - 1 = 8
    - You write: **45678** *(Notice: 54,321 + 45,678 = 99,999)*

7. **Step 7: Add all 5 numbers to reveal the total.**

```
  12345  (Volunteer's 1st Number)
  23456  (Volunteer's 2nd Number)
  76543  (Your 3rd Number)
  54321  (Volunteer's 4th Number)
+ 45678  (Your 5th Number)
——————————————————————————————
 212343  (Matches Prediction Exactly!)
```

The Math Behind the Trick

The trick works because the **"Make 9" numbers always form pairs that add to 99,999**.

Why It Works

For example:

23,456 + 76,543 = 99,999

Each matching pair of digits adds to **9**:

2 + 7 = 9
3 + 6 = 9
4 + 5 = 9
5 + 4 = 9
6 + 3 = 9

The second pair works the same way:

54,321 + 45,678 = 99,999

So the four numbers after the first always give:

99,999 + 99,999 = 199,998

Why You Can Predict the Total

You only need to see the first number.

For example:

12,345 + 199,998 = 212,343

So you can write **212,343** as your prediction before the other four numbers are written.

Remember

Make 9 → 99,999
Two pairs → 199,998
First Number + 199,998 → Final Total

That is the simple math behind the prediction.

Magic Trick 2: The 1089 Trick

The Effect

The spectator chooses a **3-digit number with different first and last digits**.

After a few simple operations, the result becomes:

1089

Starting Condition

Choose any **3-digit number** where the **first and last digits are different**.

For example:
532 ✅
211 ✅
111 ❌

Example: 532

Step 1: Reverse the number.
532 → 235

Step 2: Subtract the smaller number from the larger.
532 - 235 = 297

Step 3: Reverse the result.
297 → 792

Step 4: Add the two numbers.
297 + 792 = 1089

Final Result
1089

Important

If the subtraction gives a result with fewer than three digits, keep the leading zero.

For example: 099

must stay: 099

before reversing it to: 990

The leading zero is part of the calculation.

Magic Trick 3: The 1001 Repetition Trick

The Effect

A spectator chooses a **3-digit number**.

Multiply it by **7, then 11, then 13**, and the number appears twice.

Example: 345

Step 1:

7 × 11 × 13 = 1001

Step 2:

345 × 1001 = 345000 + 345 = 345345

Final Result

345345

The number: 345

has become:

345 | 345

Performance Idea

You can frame this as a prediction:

“Choose any 3-digit number. Multiply it through these three simple numbers and watch what happens.”

The spectator does the multiplication and discovers the repetition themselves.

Magic Trick 4: The 6174 Trick

The Effect

Start with a **4-digit number that has at least two different digits**.

Arrange the digits from largest to smallest.
Then arrange them from smallest to largest.
Subtract.
Repeat.

The process reaches: **6174**

Important

Choose a 4-digit number with **at least two different digits**.

For example:
3524 ✅
2025 ✅
1222 ✅
1111 ❌

Keep four digit places during the process.

For example: **0378**

should not be shortened to: **378**

Example: Start With 3524

Step 1: Arrange descending and ascending.

5432
2345

Subtract:
5432 - 2345 = 3087

Step 2: Repeat.

8730

0378

Subtract:

8730 - 0378 = 8352

Step 3: Repeat again.

8532

2358

Subtract:

8532 - 2358 = 6174

Final Result

6174

The Strange Part

Once you reach 6174:

7641 - 1467 = 6174

So the same number comes back again.

Performance Tip

Let the spectator choose the starting number.

Do not tell them the destination.

Let them perform the process repeatedly until 6174 appears.

Magic Trick 5: Pocket Change Prediction

The Effect

Ask someone for their **age** and the number of **cents** in their pocket.
After a few simple calculations, the final number will contain both pieces of information:

Age | Cents

For example: **2347**
Means: **23 years | 47 cents**

Important

The cents should be treated as a **two-digit number from 00 to 99**.

For example:

47 cents → 47
7 cents → 07

This keeps the cents in the final **two digits**.

Example

Age = 23
Cents = 47

Step 1: Double the age.

23 × 2 = 46

Step 2: Add 5.

46 + 5 = 51

Step 3: Multiply by 50.

51 × 50 = 2550

Step 4: Add the cents.

2550 + 47 = 2597

Step 5: Subtract 250.

2597 - 250 = **2347**

Step 6: Split the final number into age and cents.

2347
→ **23 | 47**

So: Age = **23**
Cents = **47**

Final Result

2347

reveals:

23 years and 47 cents

Magic Trick 6: Reveal the Date of Birth

The Effect

You can reveal three parts of someone's birth date using a simple number calculation:

Birth Month | Birth Date | Last Two Digits of Birth Year

For example:

12 | 31 | 99

This means:

December | 31 | Year ending in 99

The calculation reveals the **month, date, and last two digits of the year**. It does not reveal the century, so **1999** and **2099** would look the same.

How to Perform the Trick

Tell the volunteer to follow **every step in order**. They should use their own birth details while you guide them through the calculation.

For the example below, we will use:

Birth Date: December 31, 1999

So:

Birth Month = 12
Birth Date = 31
Last Two Digits of Year = 99

Step 1: Start With the Birth Month

Ask the volunteer for their **birth month number**.

January = 1
February = 2
March = 3
...
December = 12

For our example:

Birth Month = 12

Write:

12

Step 2: Add 18

Take the month number and add **18**.

12 + 18 = 30

Your new number is:

30

Step 3: Multiply by 25

Take the number you just got and multiply it by **25**.

30 × 25 = 750

Your new number is:

750

Step 4: Subtract 333

Now subtract **333** from the result.

750 − 333 = 417

Your new number is:

417

Step 5: Multiply by 8

Multiply the new number by **8**.

417 × 8 = 3336

Your new number is:

3336

Step 6: Subtract 554

Now subtract **554**.

3336 − 554 = 2782

Your new number is:

2782

Step 7: Divide by 2

Divide the result by **2**.

2782 ÷ 2 = 1391

Your new number is:

1391

Step 8: Add the Birth Date

Now ask the volunteer to take the **day of the month they were born on**.

For our example, the birth date is **31**.

Add it to the current number:

1391 + 31 = 1422

Your new number is:

1422

Step 9: Multiply by 5

Multiply the result by **5**.

1422 × 5 = 7110

Your new number is:

7110

Step 10: Add 692

Now add **692**.

7110 + 692 = 7802

Your new number is:

7802

Step 11: Multiply by 20

Multiply the result by **20**.

7802 × 20 = 156040

Your new number is:

156040

Step 12: Add the Last Two Digits of the Birth Year

Now ask the volunteer for the **last two digits of their birth year**.

For 1999, the last two digits are:

99

Add them to the current number:

156040 + 99 = 156139

Your new number is:

156139

Step 13: Subtract 32940

Finally, subtract **32940**.

156139 − 32940 = 123199

You now have the final number:

123199

Step 14: Read the Final Number

Now split the final number into **three parts**:

12 | 31 | 99

Read the three parts as:

12 = Birth Month
31 = Birth Date
99 = Last Two Digits of Birth Year

So the birth date is:

December 31 | Year ending in 99

The full year cannot be determined from the calculation alone because the calculation only reveals the **last two digits**.

Another Example

Suppose someone was born on:

5 March 2007

Use:

Birth Month = 3
Birth Date = 5
Last Two Digits of Year = 07

Follow the same steps in exactly the same order.

The final result is:

30507

Now separate it into three two-digit parts:

03 | 05 | 07

Read it as:

March | 5 | Year ending in 07

So the calculation reveals:

March 5 | Year ending in 07

Since this example person's full year is already known to be **2007**, the complete date is:

March 5, 2007

Important

When reading the final result, treat the month, date, and year as **two-digit numbers**.

For example:

1 January 2007 → 01 | 01 | 07
5 March 2007 → 03 | 05 | 07
12 December 1999 → 12 | 12 | 99

This makes the final number easy to read as:

MM | DD | YY
